INORGANIC CUME
November 13, 2015

Please answer the questions as best as you can. Show your work where relevant, and write down your logic even if you cannot complete the question. Partial credit will be given where earned.

1. (20) In a few sentences each, describe the following techniques and what information you can get out of them. You may use diagrams as you see relevant.
   (a) X-ray diffraction
   (b) X-ray photoelectron spectroscopy
   (c) Raman spectroscopy
   (d) Which region of the material (surface or bulk) are (a) and (b) sensitive to?

2. (10) Excitons in solids
   (a) What is an exciton?
   (b) Excitons can exist in semiconductors and in polymers, but their lifetimes in polymers are orders of magnitude longer. Why is this?

3. (15) Au-Si phase diagram: Figure 1 is the phase diagram for Au-Si alloys as a function of temperature and weight % of Si in the alloy.
   (a) What are the melting points of pure Au and pure Si?
   (b) What is the definition of the eutectic point?
   (c) What is the eutectic point of the Au-Si alloy?

4. (15) Nanocrystals and ligand exchange
   (a) Name 2 roles that terminal ligands play for chemically synthesized nanoparticles (either in the synthesis or in the utilization of nanoparticles)? A relevant example is thiol groups on an Au nanocrystal.
   (b) Name 2 methods could be used to study ligand exchange (in situ or otherwise)? For each, describe what information would be learned from the measurements.

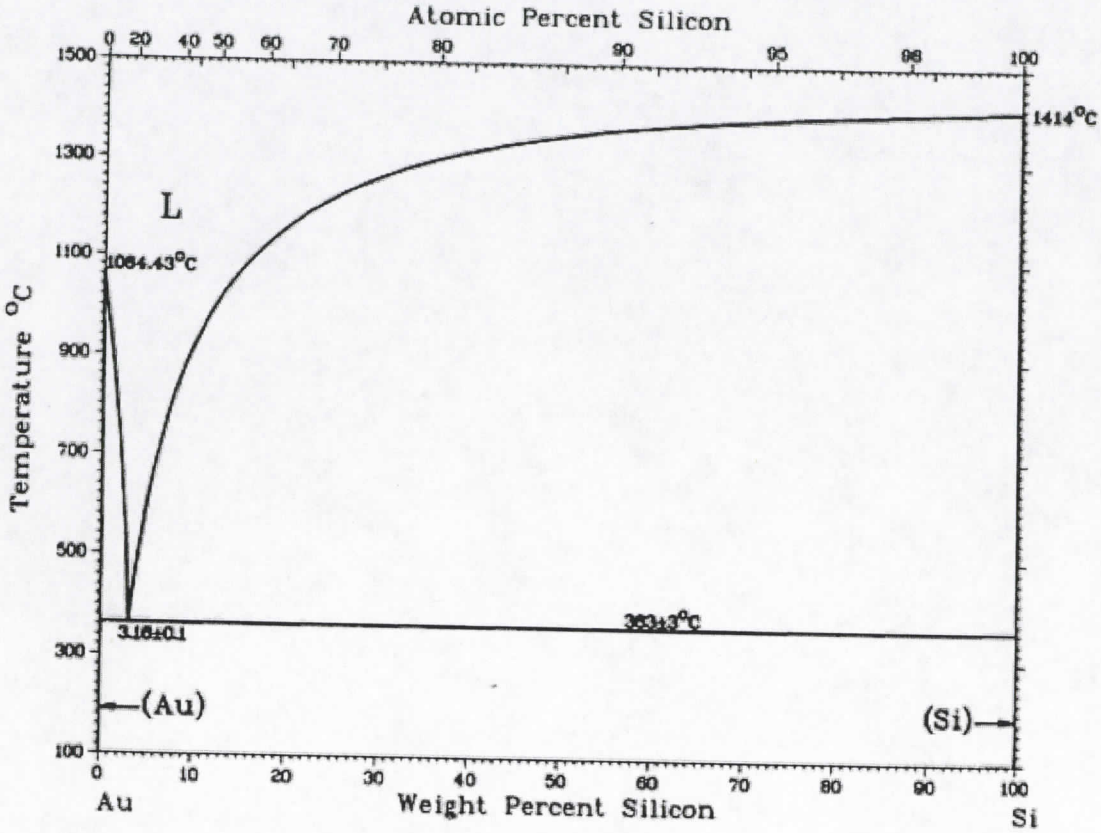

Figure 1: Au-Si alloy phase diagram, from Okamoto and Massalski, 1987.